ELL8299/COL8383/COL873: Trustworthy Large Language Models

Quiz 3 (Semester I, 2026-27)

Answer the questions in the spaces provided. No extra pages will be given.

Name: _____ Entry Number: _____

Total Marks: 25 **Time: 50 Minutes** **26/10/2026**

1. Local bias is defined as the divergence between next-token distributions after two counterfactual contexts. Context c_1 is "The man performing surgery is a _____" and c_2 is "The woman performing surgery is a _____". Over the set { doctor, nurse, teacher }, we have $p_1 = (0.5, 0.3, 0.2)$ and $p_2 = (0.3, 0.5, 0.2)$ where p_1 and p_2 denote the two probability distributions over the possible next words, conditioned on the two counterfactual contexts c_1 and c_2 respectively. Use natural logarithms for all entropy and divergence calculations in this question.

   (a) Compute $D_{KL}(p_1 \parallel p_2)$. [1]

   (b) Compute the squared Hellinger distance $H^2 = 1 - \sum \sqrt{p_1 p_2}$. [1]

2. An answer consists of three claims with uncertainty scores $u = (0.05, 0.10, 0.80)$, policy-support verdicts $v = (S, S, C)$ where S = Supported and C = Contradicted, and importance weights $w = (0.2, 0.3, 0.5)$. Claim 3 is marked as decision-critical.

   (a) Compute the values of the following metrics: R_{avg} (Average Risk), R_{max} (Maximum Risk), G_{avg} (The fraction of claims that are contradicted), G_w (Weighted Contradiction Rate). [2]

   (b) Based on these metrics and the presence of a decision-critical claim, state whether the system should provide the answer or abstain. Justify your choice. [1]

3. HELM measures demographic representation as the Total Variation Distance (TVD) between the observed group-word distribution and a uniform reference.

   (a) The generations contain 6 male and 2 female terms. Compute the TVD. [1]

(b) Race, with three groups: counts are Asian: 10, Hispanic: 0, White: 0. Compute the TVD. [1]

4. Six sampled answers $a_{1:6}$ are compared with an Natural Language Inference (NLI) model. Every pair among a_1, a_2, a_3, a_4 entail each other in both directions. a_5 entails a_1, but a_1 does not entail a_5. a_6 has no entailment with any other answer.

   (a) Form the clusters using mutual entailment. [1]

   (b) Compute the empirical semantic entropy. [1]

   (c) What would naive entropy over the six distinct strings be? [1]

5. OCR correctly reads both rows of a routing table: "B.Tech → course instructor → Form A" and "Ph.D. → programme coordinator → Form P". The assistant tells a Ph.D. student to submit Form A to the instructor. Which of the three steps failed: read, select or bind? Why wouldn't OCR confidence catch it? [2]

6. The prompt is "After retirement, I plan to move to _____", with $p = (\text{Goa: } 0.5, \text{Vizag: } 0.15, \text{Shimla: } 0.3, \text{Munnar: } 0.05)$. The 3-layer scores are as follows:

Token	$\mathbf{g}_1$	$\mathbf{g}_2$	$\mathbf{g}_3$
Goa	1	0	1
Vizag	0	0	1
Shimla	0	1	0
Munnar	1	0	0

   Eight candidates are drawn in this order: Munnar, Goa, Shimla, Goa, Vizag, Shimla, Goa, Goa. Layer 1 pairs them (1,2), (3,4), (5,6), (7,8). Each later layer pairs the winners in bracket order. Use the tie rule "ties go to the left element".

   (a) Trace every match in all three layers and give the output token. [2]

(b) Consider a 1-layer, 2-candidate tournament in which ties are broken by a fair coin. Prove that such a tournament is distortion-free. [2]

(c) Prove that the L-layer, 2-candidate tournament is distortion-free for every L. State exactly where your proof uses the assumption that scores are independent across layers. [2]

7. The Kirchenbauer-Geiping-Wen (KGW) algorithm achieves white-box watermarking by distorting the LLM's natural logit distribution. Based on this algorithm, state the exact piecewise mathematical formula for the watermarked probability $\hat{p}_k^{(t)}$ of a token k, distinguishing between the case where k is in the "green list" G and when k is in the "red list" R. Include the hardness parameter and the unwatermarked logits $l^{(t)}$. [2]

8. Match each observation below to exactly one term from the following list. Use each term exactly once: *exposure bias, hallucination snowballing, sycophancy, long-tailed knowledge, outdated knowledge.* [5]

(a) A model answers correctly at first, but changes to an incorrect answer after the user insists that the incorrect answer is right. _____

(b) A model reports the office-holder at its training cutoff as the current office-holder, although a replacement took office after the cutoff. _____

(c) During teacher-forced training, a model conditions on ground-truth prefixes; during generation, it conditions on its own potentially erroneous outputs. _____

(d) A model commits to an incorrect "yes" answer, then generates additional false claims to justify that initial answer. _____

(e) A fact appears in very few pretraining documents, and the model struggles to answer questions about it compared with frequently represented facts. _____

— Space for Rough Work —